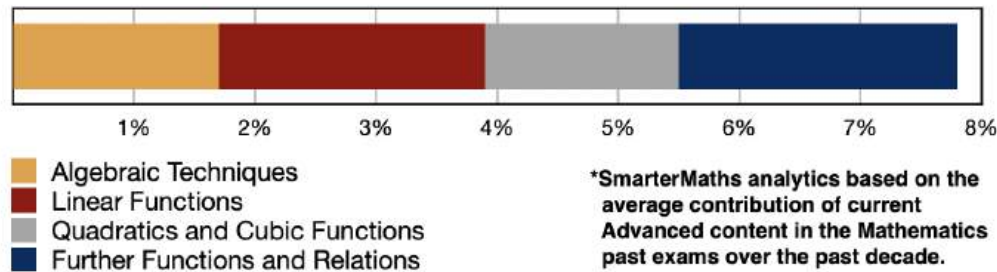


ADV: Functions (Adv), F1 Working with Functions (Adv)
Further Functions and Relations (Y11)
Composite Functions (Y11)

Teacher: Ljiljana Todic

Exam Equivalent Time: 145.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

F1 Working With Functions



HISTORICAL CONTRIBUTION

- *F1 Working with Functions* is a Year 11 topic whose content represents the lowest of low hanging fruit in the new *Advanced* course.
- *F1 Working with Functions* includes new and a significant re-categorisation of old syllabus content.
- We have split the topic into 5 categories for analysis purposes: 1-*Algebraic Techniques*, 2-*Linear Functions*, 3-*Quadratics and Cubic Functions*, 4-*Composite Functions* and 5-*Further Functions and Relations*.
- This analysis looks at *Composite Functions*.

HSC ANALYSIS - What to expect and common pitfalls

- *Composite Functions* is brand new content that is therefore absent from the "past contributions" bar chart above.
- Our analysis has found composite functions to be a highly examinable topic area. By this we mean it lends itself to a broad spectrum of questions at varying degrees of difficulty - which has been the experience of other States with an exam history in this topic.
- We recommend a close revision of examples where students need to state the domain and range of composite functions. Numerous examples in the database look at this.

Questions

1. Functions, 2ADV F1 2019 HSC 2 MC

What values of x satisfy $4 - 3x \leq 12$?

- (A) $x \leq -\frac{16}{3}$
 (B) $x \geq -\frac{16}{3}$
 (C) $x \leq -\frac{8}{3}$
 (D) $x \geq -\frac{8}{3}$

2. Functions, 2ADV F1 SM-Bank 2 MC

Let $f(x)$ and $g(x)$ be functions such that $f(2) = 5$, $f(3) = 4$, $g(2) = 5$, $g(3) = 2$ and $g(4) = 1$.

The value of $f(g(3))$ is

- A. 1
 B. 2
 C. 4
 D. 5

3. Functions, 2ADV F1 SM-Bank 5 MC

Let $g(x) = x^2 + 2x - 3$ and $f(x) = e^{2x+3}$.

Then $f(g(x))$ is given by

- A. $e^{4x+6} + 2e^{2x+3} - 3$
 B. $2x^2 + 4x - 6$
 C. e^{2x^2+4x-3}
 D. e^{2x^2+4x-6}

4. Functions, 2ADV F1 SM-Bank 9 MC

If $f(x) = \frac{1}{2}e^{3x}$ and $g(x) = \log_e(2x) + 3$ then $g(f(x))$ is equal to

- A. $3(x + 1)$
- B. $e^{3x} + 3$
- C. e^{8x+9}
- D. $\log_e(3x) + 3$

5. Functions, 2ADV F1 SM-Bank 13 MC

Which one of the following functions satisfies the functional equation $f(f(x)) = x$?

- A. $f(x) = 2 - x$
- B. $f(x) = x^2$
- C. $f(x) = 2\sqrt{x}$
- D. $f(x) = x - 2$

6. Functions, 2ADV F1 2013 HSC 3 MC

Which inequality defines the domain of the function $f(x) = \frac{1}{\sqrt{x+3}}$?

- (A) $x > -3$
- (B) $x \geq -3$
- (C) $x < -3$
- (D) $x \leq -3$

7. Functions, 2ADV F1 2020 HSC 1 MC

Which inequality gives the domain of $y = \sqrt{2x-3}$?

- A. $x < \frac{3}{2}$
- B. $x > \frac{3}{2}$
- C. $x \leq \frac{3}{2}$
- D. $x \geq \frac{3}{2}$

8. Functions, 2ADV F1 SM-Bank 8 MC

Let $f(x) = x^2$

Which one of the following is **not** true?

- A. $f(xy) = f(x)f(y)$
- B. $f(x) - f(-x) = 0$
- C. $f(2x) = 4f(x)$
- D. $f(x - y) = f(x) - f(y)$

9. Functions, 2ADV F1 SM-Bank 12 MC

If $f(x - 1) = x^2 - 2x + 3$, then $f(x)$ is equal to

- A. $x^2 - 2$
- B. $x^2 + 2$
- C. $x^2 - 2x + 4$
- D. $x^2 - 4x + 6$

10. Functions, 2ADV F1 SM-Bank 15 MC

If the equation $f(2x) - 2f(x) = 0$ is true for all real values of x , then $f(x)$ could equal

- A. $\frac{x^2}{2}$
- B. $\sqrt{2x}$
- C. $2x$
- D. $x - 2$

11. Functions, 2ADV F1 SM-Bank 21 MC

A circle with centre $(a, -2)$ and radius 5 units has equation

$$x^2 - 6x + y^2 + 4y = b \text{ where } a \text{ and } b \text{ are real constants.}$$

The values of a and b are respectively

- A. -3 and 38
- B. 3 and 12
- C. -3 and -8
- D. 3 and 18

12. Functions, 2ADV F1 SM-Bank 1 MC

Let $h(x) = \frac{1}{x-1}$ for $-1 < h < 1$.

Which one of the following statements about h is **not** true?

- A. $h(x)h(-x) = -h(x^2)$
 - B. $h(x) - h(0) = xh(x)$
 - C. $h(x) - h(-x) = 2xh(x^2)$
 - D. $(h(x))^2 = h(x^2)$
-

13. Functions, 2ADV F1 SM-Bank 4 MC

The function $f(x)$ satisfies the functional equation $f(f(x)) = x$ for $\{x: \text{all } x, x \neq 1\}$.

The rule for the function is

- A. $f(x) = x + 1$
 - B. $f(x) = x - 1$
 - C. $f(x) = \frac{x-1}{x+1}$
 - D. $f(x) = \frac{x+1}{x-1}$
-

14. Functions, 2ADV F1 SM-Bank 6 MC

Let $f(x) = e^x + e^{-x}$.

$f(2u)$ is equal to

- A. $f(u) + f(-u)$
 - B. $2f(u)$
 - C. $(f(u))^2 - 2$
 - D. $(f(u))^2 + 2$
-

15. Functions, 2ADV F1 SM-Bank 14 MC

Let $g(x) = \log_2(x)$, $x > 0$

Which one of the following equations is true for all positive real values of x ?

- A. $2g(8x) = g(x^2) + 8$
 - B. $2g(8x) = g(x^2) + 6$
 - C. $2g(8x) = (g(x) + 8)^2$
 - D. $2g(8x) = g(2x) + 6$
-

16. Functions, 2ADV F1 2007 HSC 1b

Solve $2x - 5 > -3$ and graph the solution on a number line. (2 marks)

17. Functions, 2ADV F1 2009 HSC 1c

Solve $|x + 1| = 5$. (2 marks)

18. Functions, 2ADV F1 2010 HSC 1d

Solve $|2x + 3| = 9$. (2 marks)

19. Functions, 2ADV F1 2006 HSC 1e

Solve $3 - 5x \leq 2$. (2 marks)

20. Functions, 2ADV F1 2011 HSC 1e

Solve $2 - 3x \leq 8$. (2 marks)

21. Functions, 2ADV F1 2017 HSC 11g

Solve $|3x - 1| = 2$. (2 marks)

22. Functions, 2ADV F1 2018 HSC 11b

Solve $1 - 3x > 10$. (2 marks)

23. Functions, 2ADV F1 2008 HSC 1d

Solve $|4x - 3| = 7$. (2 marks)

24. Functions, 2ADV F1 SM-Bank 10

Let $f(x) = x^2 + 1$ and $g(x) = 2x + 1$. Write down the rule of $f(g(x))$. (1 mark)

25. Functions, 2ADV F1 SM-Bank 33

- State the domain and range of $y = -\sqrt{12 - x^2}$. (2 marks)
- Sketch the graph. (1 mark)

26. Functions, 2ADV F1 SM-Bank 41

Find the values of x for which $|x + 1| = 5$. (2 marks)

27. Functions, 2ADV F1 SM-Bank 42

Find the values of x for which $|x - 3| = 1$. (2 marks)

28. Functions, 2ADV F1 SM-Bank 44

Solve $|x - 2| = 3$. (2 marks)

29. Functions, 2ADV F1 SM-Bank 37

Find all values of x for which $|x - 4| = \frac{x}{2} + 7$. (3 marks)

30. Functions, 2ADV F1 2016 HSC 11a

Sketch the graph of $(x - 3)^2 + (y + 2)^2 = 4$. (2 marks)

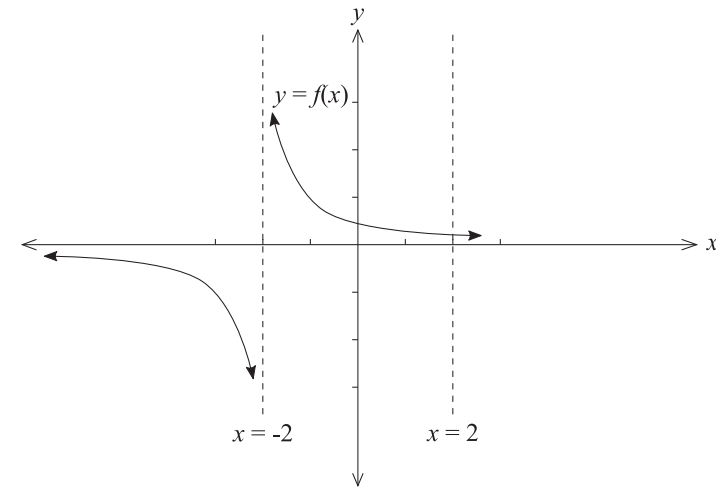
31. Functions, 2ADV F1 2019 MET1-N 2

Let $f(x) = -x^2 + x + 4$ and $g(x) = x^2 - 2$.

- Find $g(f(3))$. (2 marks)
- Express $f(g(x))$ in the form $ax^4 + bx^2 + c$, where a , b and c are non-zero integers. (2 marks)

32. Functions, 2ADV F1 SM-Bank 36

Consider the function $f(x) = \frac{1}{x + 2}$.



- Sketch the graph $y = f(-x)$. (2 marks)
- On the same graph, sketch $y = -f(x)$. (2 marks)

33. Functions, 2ADV F1 SM-Bank 30

Given $f(x) = \sqrt{x}$ and $g(x) = 25 - x^2$

- Find $g(f(x))$. (1 mark)
- Find the domain and range of $f(g(x))$. (2 marks)

34. Functions, 2ADV F1 2019 HSC 13e

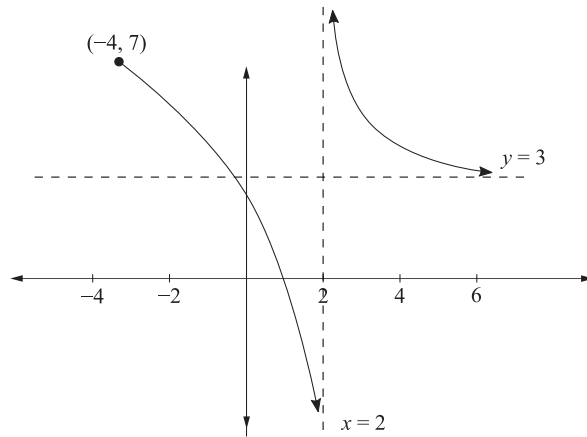
- Sketch the graph of $y = |x - 1|$ for $-4 \leq x \leq 4$. (1 mark)
- Using the sketch from part i, or otherwise, solve $|x - 1| = 2x + 4$. (2 marks)

35. Functions, 2ADV F1 EQ-Bank 11

Given the function $f(x) = \sqrt{3 - x}$ and $g(x) = x^2 - 2$, sketch $y = g(f(x))$ over its natural domain. (2 marks)

36. Functions, 2ADV F1 EQ-Bank 6

The graph of $f(x)$ is shown below. It has asymptotes at $y = 3$ and $x = 2$.



Using interval notation, state the domain and range of $f(x)$. (2 marks)

37. Functions, 2ADV F1 EQ-Bank 7

The current of an electrical circuit, measured in amps (A), varies inversely with its resistance, measured in ohms (R).

When the resistance of a circuit is 28 ohms, the current is 3 amps.

What is the current when the resistance is 8 ohms? (2 marks)

38. Functions, 2ADV F1 EQ-Bank 8

Jacques is a marine biologist and finds that the mass of a crab is directly proportional to the cube of the diameter of its shell.

If a crab with a shell diameter of 15 cm weighs 680 grams, what will be the diameter of a crab that weighs 1.1 kilograms? Give your answer to 1 decimal place. (2 marks)

39. Functions, 2ADV F1 EQ-Bank 26

Fuifui finds that for Giant moray eels, the mass of an eel is directly proportional to the cube of its length.

An eel of this species has a length of 25 cm and a mass of 4350 grams.

What is the expected length of a Giant moray eel with a mass of 6.2 kg? Give your answer to one decimal place. (3 marks)

40. Functions, 2ADV F1 EQ-Bank 27

The stopping distance of a car on a certain road, once the brakes are applied, is directly proportional to the square of the speed of the car when the brakes are first applied.

A car travelling at 70 km/h takes 58.8 metres to stop.

How far does it take to stop if it is travelling at 105 km/h? (3 marks)

41. Functions, 2ADV F1 SM-Bank 32

Find the centre and radius of the circle with the equation

$$x^2 - 12x + y^2 + 2y - 12 = 0 \quad (2 \text{ marks})$$

42. Functions, 2ADV F1 SM-Bank 31

Find the domain and range of $f(g(x))$ given

$$f(x) = 2x^2 - 8x \text{ and } g(x) = x + 2. \quad (2 \text{ marks})$$

43. Functions, 2ADV F1 2010 HSC 1c

Write down the equation of the circle with centre $(-1, 2)$ and radius 5. (1 mark)

44. Functions, 2ADV F1 2010 HSC 1g

Let $f(x) = \sqrt{x - 8}$. What is the domain of $f(x)$? (1 mark)

45. Functions, 2ADV F1 2017 HSC 11h

Find the domain of the function $f(x) = \sqrt{3 - x}$. (2 marks)

46. Functions, 2ADV F1 SM-Bank 3

Let $f(x) = \sqrt{x + 1}$ for $x \geq 0$

i. State the range of $f(x)$. (1 mark)

ii. Let $g(x) = x^2 + 4x + 3$, where $x \leq c$ and $c \leq 0$.

Find the largest possible value of c such that the range of $g(x)$ is a subset of the domain of $f(x)$. (2 marks)

47. Functions, 2ADV F1 SM-Bank 7

Let $f(x) = \log_e(x)$ for $x > 0$, and $g(x) = x^2 + 1$ for all x .

i. Find $h(x)$, where $h(x) = f(g(x))$. (1 mark)

ii. State the domain and range of $h(x)$. (2 marks)

iii. Show that $h(x) + h(-x) = f((g(x))^2)$. (2 marks)

48. Functions, 2ADV F1 2020 HSC 24

The circle of $x^2 - 6x + y^2 + 4y - 3 = 0$ is reflected in the x -axis.

Sketch the reflected circle, showing the coordinates of the centre and the radius. (3 marks)

49. Functions, 2ADV F1 SM-Bank 11

Given $f(x) = \sqrt{x^2 - 9}$ and $g(x) = x + 5$

a. Find integers c and d such that $f(g(x)) = \sqrt{(x+c)(x+d)}$ (2 marks)

b. State the domain for which $f(g(x))$ is defined. (2 marks)

Worked Solutions

1. Functions, 2ADV F1 2019 HSC 2 MC

$$4 - 3x \leq 12$$

$$-3x \leq 8$$

$$x \geq -\frac{8}{3}$$

$$\Rightarrow D$$

2. Functions, 2ADV F1 SM-Bank 2 MC

$$f(g(3)) = f(2)$$

$$= 5$$

$$\Rightarrow D$$

3. Functions, 2ADV F1 SM-Bank 5 MC

By trial and error,

$$\text{Consider: } f(x) = e^{2x^2+4x-3}$$

$$f(g(x)) = e^{2(x^2+2x-3)+3}$$

$$= e^{2x^2+4x-3}$$

$$\Rightarrow C$$

4. Functions, 2ADV F1 SM-Bank 9 MC

$$g(f(x)) = \log_e \left(2 \times \frac{1}{2} e^{3x} \right) + 3$$

$$= \log_e e^{3x} + 3$$

$$= 3x + 3$$

$$= 3(x + 1)$$

$$\Rightarrow A$$

5. Functions, 2ADV F1 SM-Bank 13 MC

By trial and error,

Consider: $f(x) = 2 - x$

$$\begin{aligned} f(f(x)) &= 2 - (2 - x) \\ &= x \end{aligned}$$

$\Rightarrow A$

6. Functions, 2ADV F1 2013 HSC 3 MC

Given $f(x) = \frac{1}{\sqrt{x+3}}$

We know $(x+3) > 0$
 $x > -3$

$\therefore$ The domain of $f(x)$ is $f(x) > -3$

$\Rightarrow A$

7. Functions, 2ADV F1 2020 HSC 1 MC

Domain exists when:

$$2x - 3 \geq 0$$

$$2x \geq 3$$

$$x \geq \frac{3}{2}$$

$\Rightarrow D$

8. Functions, 2ADV F1 SM-Bank 8 MC

By trial and error,

Consider option D :

$$\begin{aligned} f(x-y) &= (x-y)^2 \\ &= x^2 - 2xy + y^2 \end{aligned}$$

$$f(x) - f(y) = x^2 - y^2$$

$$\therefore f(x-y) \neq f(x) - f(y)$$

$\Rightarrow D$

9. Functions, 2ADV F1 SM-Bank 12 MC

Let $g(x) = f(x-1)$

$$g(x+1) = f(x)$$

$$\begin{aligned} g(x+1) &= (x+1)^2 - 2(x+1) + 3 \\ &= x^2 + 2x + 1 - 2x - 2 + 3 \\ &= x^2 + 2 \end{aligned}$$

$\Rightarrow B$

10. Functions, 2ADV F1 SM-Bank 15 MC

We need $f(2x) = 2f(x)$,

Consider C ,

$$f(x) = 2x,$$

$$f(2x) = 2(2x)$$

$$= 2f(x)$$

$\Rightarrow C$

11. Functions, 2ADV F1 SM-Bank 21 MC

$$x^2 - 6x + y^2 + 4y = b$$

Completing the squares:

$$x^2 - 6x + 3^2 - 9 + y^2 + 4y + 2^2 - 4 = b$$

$$(x - 3)^2 + (y + 2)^2 - 13 = b$$

$$(x - 3)^2 + (y + 2)^2 = b + 13$$

$$\therefore a = 3$$

$$\therefore b + 13 = 25 \Rightarrow b = 12$$

$$\Rightarrow B$$

12. Functions, 2ADV F1 SM-Bank 1 MC

By trial and error, consider option *E*:

$$h(x) = \frac{1}{x - 1}$$

$$(h(x))^2 = \frac{1}{(x - 1)^2} = \frac{1}{x^2 - 2x + 1}$$

$$h(x^2) = \frac{1}{x^2 - 1} \neq (h(x))^2$$

$$\Rightarrow D$$

13. Functions, 2ADV F1 SM-Bank 4 MC

By trial and error:

$$\text{Consider } f(x) = \frac{x + 1}{x - 1}$$

$$f(f(x)) = \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1}$$

$$= \frac{x + 1 + x - 1}{x + 1 - x + 1}$$

$$= x$$

$$\Rightarrow D$$

14. Functions, 2ADV F1 SM-Bank 6 MC

By trial and error,

$$\text{Consider } (f(u))^2 - 2:$$

$$f(2u) = e^{2u} + e^{-2u}$$

$$\begin{aligned} (f(u))^2 &= (e^u + e^{-u})^2 \\ &= e^{2u} + 2 + e^{-2u} \end{aligned}$$

$$\therefore f(2u) = (f(u))^2 - 2$$

$$\Rightarrow C$$

♦ Mean mark 44%.

15. Functions, 2ADV F1 SM-Bank 14 MC

Consider Option *B*:

$$\text{LHS} = 2g(8x)$$

$$= 2 \log_2(8x)$$

$$= 2 \log_2(8) + 2 \log_2(x)$$

$$= 2 \log_2(2^3) + 2 \log_2(x)$$

$$= 6 + \log_2(x^2)$$

$$= g(x^2) + 6$$

$$\Rightarrow B$$

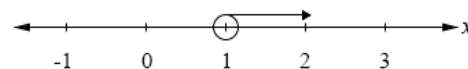
♦♦ Mean mark 35%.

16. Functions, 2ADV F1 2007 HSC 1b

$$2x - 5 > -3$$

$$2x > 2$$

$$x > 1$$



17. Functions, 2ADV F1 2009 HSC 1c

$$|x + 1| = 5$$

$$(x + 1) = 5 \quad - (x + 1) = 5$$

$$x = 4 \quad -x - 1 = 5$$

$$x = -6$$

$$\therefore x = 4 \text{ or } -6$$

18. Functions, 2ADV F1 2010 HSC 1d

$$|2x + 3| = 9$$

$$2x + 3 = 9 \quad - (2x + 3) = 9$$

$$2x = 6 \quad -2x - 3 = 9$$

$$x = 3 \quad -2x = 12$$

$$x = -6$$

$$\therefore x = 3 \text{ or } -6$$

19. Functions, 2ADV F1 2006 HSC 1e

$$3 - 5x \leq 2$$

$$-5x \leq -1$$

$$x \geq \frac{1}{5}$$

20. Functions, 2ADV F1 2011 HSC 1e

$$2 - 3x \leq 8$$

$$-3x \leq 6$$

$$x \geq -\frac{6}{3}$$

$$x \geq -2$$

21. Functions, 2ADV F1 2017 HSC 11g

$$|3x - 1| = 2$$

$$3x - 1 = 2 \quad \text{or} \quad - (3x - 1) = 2$$

$$x = 1 \quad -3x = 1$$

$$x = -\frac{1}{3}$$

$$\therefore x = 1 \text{ or } -\frac{1}{3}$$

MARKER'S COMMENT: Note that both conditions must be satisfied! Dealing with negative signs and division for inequalities produced many errors.

22. Functions, 2ADV F1 2018 HSC 11b

$$1 - 3x > 10$$

$$-3x > 9$$

$$x < -3$$

23. Functions, 2ADV F1 2008 HSC 1d

$$|4x - 3| = 7$$

$$4x - 3 = 7 \quad - (4x - 3) = 7$$

$$4x = 10 \quad -4x + 3 = 7$$

$$x = \frac{5}{2} \quad -4x = 4$$

$$x = -1$$

$$\therefore x = \frac{5}{2} \text{ or } -1$$

24. Functions, 2ADV F1 SM-Bank 10

$$f(g(x)) = f(2x + 1)$$

$$= (2x + 1)^2 + 1$$

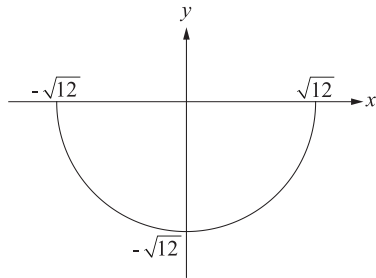
25. Functions, 2ADV F1 SM-Bank 33

i. $y = -\sqrt{12 - x^2}$

Domain: $-\sqrt{12} \leq x \leq \sqrt{12}$

Range: $-\sqrt{12} \leq y \leq 0$

ii.



26. Functions, 2ADV F1 SM-Bank 41

$$|x + 1| = 5$$

$$(x + 1)^2 = 5^2$$

$$x^2 + 2x + 1 = 25$$

$$x^2 + 2x - 24 = 0$$

$$(x + 6)(x - 4) = 0$$

$$\therefore x = 4 \text{ or } -6$$

27. Functions, 2ADV F1 SM-Bank 42

$$|x - 3| = 1$$

Method 1

$$(x - 3)^2 = 1$$

$$x^2 - 6x + 9 = 1$$

$$x^2 - 6x + 8 = 0$$

$$(x - 4)(x - 2) = 0$$

$$\therefore x = 2 \text{ or } 4$$

Method 2

$$(x - 3) = 1 \quad -(x - 3) = 1$$

$$x = 4 \quad -x + 3 = 1$$

$$x = -2$$

28. Functions, 2ADV F1 SM-Bank 44

$$|x - 2| = 3$$

$$(x - 2)^2 = 3^2$$

$$(x^2 - 4x + 4) = 9$$

$$x^2 - 4x - 5 = 0$$

$$(x - 5)(x + 1) = 0$$

$$\therefore x = -1 \text{ or } 5$$

29. Functions, 2ADV F1 SM-Bank 37

$$x - 4 = \frac{x}{2} + 7 \quad \text{or} \quad -(x - 4) = \frac{x}{2} + 7$$

$$2x - 8 = x + 14 \quad -2x + 8 = x + 14$$

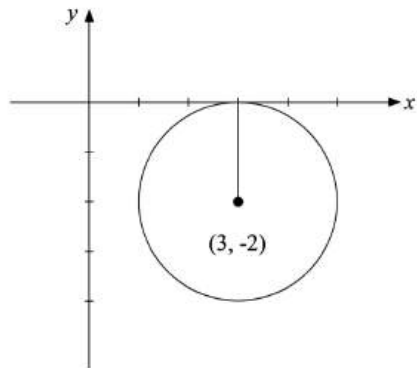
$$x = 22 \quad 3x = -6$$

$$x = -2$$

$$\therefore x = 22 \text{ or } -2$$

30. Functions, 2ADV F1 2016 HSC 11a

$(x - 3)^2 + (y + 2)^2 = 4$ is a circle,
centre $(3, -2)$, radius 2.



31. Functions, 2ADV F1 2019 MET1-N 2

a. $f(3) = -3^2 + 3 + 4$
 $= -2$

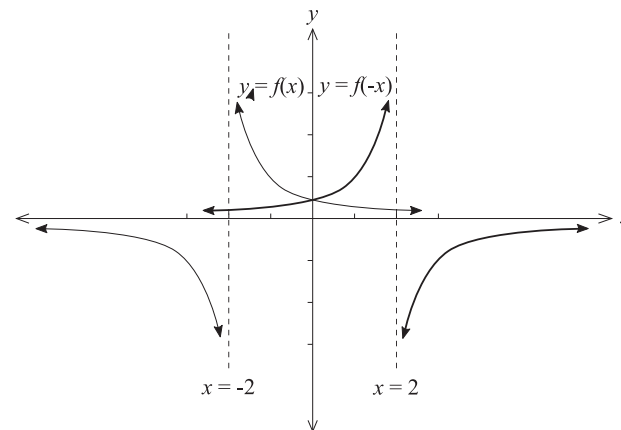
$$\begin{aligned} g(f(3)) &= g(-2) \\ &= (-2)^2 - 2 \\ &= 2 \end{aligned}$$

b. $f(g(x)) = -(x^2 - 2)^2 + (x^2 - 2) + 4$
 $= -(x^4 - 4x^2 + 4) + x^2 + 2$
 $= -x^4 + 5x^2 - 2$

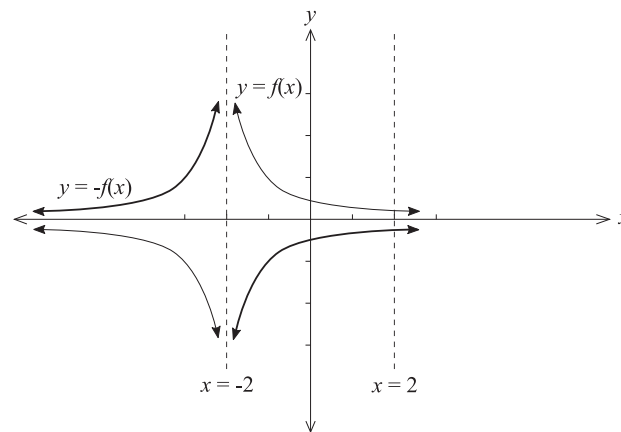
32. Functions, 2ADV F1 SM-Bank 36

i. Sketch $y = \frac{1}{x+2}$

$y = f(-x) \Rightarrow$ reflect $y = \frac{1}{x+2}$ in the y -axis.



ii. $y = -f(x) \Rightarrow$ reflect $y = \frac{1}{x+2}$ in the x -axis.



33. Functions, 2ADV F1 SM-Bank 30

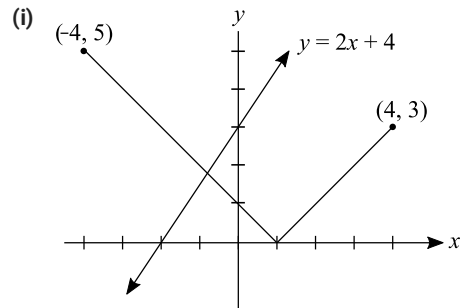
$$\begin{aligned}\text{i. } g(f(x)) &= 25 - (f(x))^2 \\ &= 25 - (\sqrt{x})^2 \\ &= 25 - x\end{aligned}$$

$$\begin{aligned}\text{ii. } f(g(x)) &= \sqrt{g(x)} \\ &= \sqrt{25 - x^2}\end{aligned}$$

$$\therefore \text{Domain: } -5 \leq x \leq 5$$

$$\therefore \text{Range: } 0 \leq y \leq 5$$

34. Functions, 2ADV F1 2019 HSC 13e



ii. By inspection, intersection when $x = -1$

Test:

$$|-1 - 1| = -2 + 4$$

$$2 = 2$$

$$\therefore \text{Intersection at } (-1, 2)$$

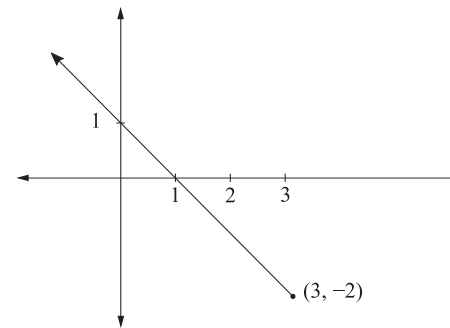
35. Functions, 2ADV F1 EQ-Bank 11

$$g(x) = x^2 - 2, \quad f(x) = \sqrt{3 - x}$$

$$\begin{aligned}g(f(x)) &= (\sqrt{3 - x})^2 - 2 \\ &= 3 - x - 2 \\ &= 1 - x\end{aligned}$$

$$\text{Since } f(x) = \sqrt{3 - x},$$

$$\Rightarrow \text{Domain: } x \leq 3$$



36. Functions, 2ADV F1 EQ-Bank 6

$$\text{Domain: } [-4, 2) \cup (2, \infty)$$

$$\text{Range: } (-\infty, \infty)$$

37. Functions, 2ADV F1 EQ-Bank 7

$$A \propto \frac{1}{R}$$

$$A = \frac{k}{R}$$

When $A = 3$, $R = 28$

$$3 = \frac{k}{28}$$

$$k = 84$$

Find A when $R = 8$:

$$\begin{aligned} A &= \frac{84}{8} \\ &= 10.5 \end{aligned}$$

38. Functions, 2ADV F1 EQ-Bank 8

$$M \propto d^3$$

$$M = kd^3$$

When $M = 680$, $d = 15$

$$680 = k \times 15^3$$

$$k = 0.201481...$$

Find d when $M = 1100$:

$$1100 = 0.20148... \times d^3$$

$$d = \sqrt[3]{\frac{1100}{0.20148...}}$$

$$= 17.608...$$

$$= 17.6 \text{ cm (to 1 d.p.)}$$

39. Functions, 2ADV F1 EQ-Bank 26

$$\text{Mass} \propto \text{length}^3$$

$$m = kl^3$$

Find k :

$$4350 = k \times 25^3$$

$$k = \frac{4350}{25^3}$$

$$= 0.2784$$

Find l when $m = 6200$:

$$6200 = 0.2784 \times l^3$$

$$l^3 = \frac{6200}{0.2784}$$

$$\therefore l = 28.13...$$

$$= 28.1 \text{ cm (to 1 d.p.)}$$

40. Functions, 2ADV F1 EQ-Bank 27

Let d = stopping distance

$$d \propto s^2$$

$$d = ks^2$$

Find k ,

$$58.8 = k \times 70^2$$

$$k = \frac{58.8}{70^2}$$

$$= 0.012$$

Find d when $s = 105$:

$$d = 0.012 \times 105^2$$

$$= 132.3 \text{ metres}$$

41. Functions, 2ADV F1 SM-Bank 32

$$x^2 - 12x + y^2 + 2y - 12 = 0$$

$$(x - 6)^2 + (y + 1)^2 - 36 - 1 - 12 = 0$$

$$(x - 6)^2 + (y + 1)^2 = 49$$

$\therefore$ Centre $(6, -1)$

$\therefore$ Radius $= 7$

42. Functions, 2ADV F1 SM-Bank 31

$$\begin{aligned} f(g(x)) &= 2(x + 2)^2 - 8(x + 2) \\ &= 2(x^2 + 4x + 4) - 8x - 16 \\ &= 2x^2 + 8x + 8 - 8x - 16 \\ &= 2(x^2 - 4) \end{aligned}$$

$\therefore$ Domain: all x

$\therefore$ Range: $-8 \leq y < \infty$

43. Functions, 2ADV F1 2010 HSC 1c

Circle with centre $(-1, 2)$, $r = 5$

$$(x + 1)^2 + (y - 2)^2 = 25$$

MARKER'S COMMENT:
Expanding this equation is not necessary!

44. Functions, 2ADV F1 2010 HSC 1g

$$f(x) = \sqrt{x - 8}$$

Domain where

$$(x - 8) \geq 0$$

$$x \geq 8$$

♦ Mean mark 49%.
MARKER'S COMMENT: $x > 8$ was a common incorrect answer.

45. Functions, 2ADV F1 2017 HSC 11h

Solution 1

Domain of $f(x) = \sqrt{3 - x}$

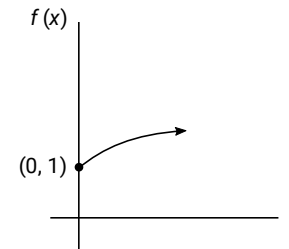
$$3 - x \geq 0$$

$$x \leq 3$$

Note domain can also be expressed as: $(-\infty, 3]$

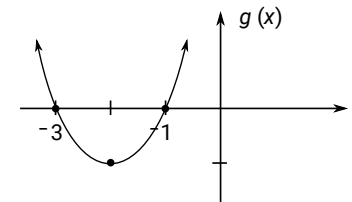
46. Functions, 2ADV F1 SM-Bank 3

i. Sketch of $f(x)$:



$\therefore$ Range: $y \geq 1$

ii. Sketch $g(x) = (x + 1)(x + 3)$



Domain of $f(x)$: $x \geq 0$

Find domain of $g(x)$ such that range $g(x)$: $y \geq 0$

Graphically, this occurs when $g(x)$ has domain:

$$x \leq -3 \text{ and } x \geq -1$$

$\therefore c = -3$

47. Functions, 2ADV F1 SM-Bank 7

$$\begin{aligned}\text{i. } h(x) &= f(x^2 + 1) \\ &= \log_e(x^2 + 1)\end{aligned}$$

$$\text{ii. Domain } (h) = \text{Domain } (g): \text{ all } x$$

$$\Rightarrow x^2 + 1 \geq 1$$

$$\Rightarrow \log_e(x^2 + 1) \geq 0$$

$$\therefore \text{Range } h(x): h \geq 0$$

$$\text{iii. LHS} = h(x) + h(-x)$$

$$= \log_e(x^2 + 1) + \log_e((-x)^2 + 1)$$

$$= \log_e(x^2 + 1) + \log_e(x^2 + 1)$$

$$= 2\log_e(x^2 + 1)$$

$$\text{RHS} = f((x^2 + 1)^2)$$

$$= 2\log_e(x^2 + 1)$$

$$\therefore h(x) + h(-x) = f((g(x))^2) \dots \text{as required}$$

♦♦ Mean mark part (a)(ii) 30%.

MARKER'S COMMENT: Many students were unsure of how to present their working in this question. Note the layout in the solution.

48. Functions, 2ADV F1 2020 HSC 24

$$x^2 - 6x + y^2 + 4y - 3 = 0$$

$$x^2 - 6x + 9 + y^2 + 4y + 4 - 16 = 0$$

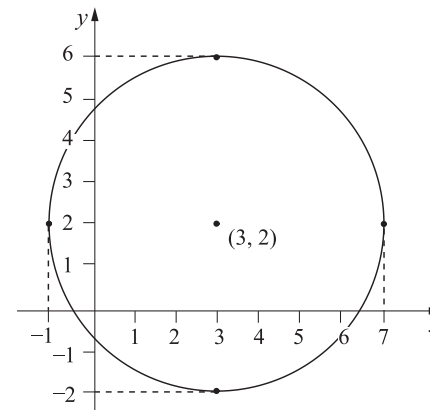
$$(x - 3)^2 + (y + 2)^2 = 16$$

$$\Rightarrow \text{Original circle has centre } (3, -2), \text{ radius} = 4$$

Reflect in x -axis:

$$\text{Centre } (3, -2) \rightarrow (3, 2)$$

♦ Mean mark 48%.

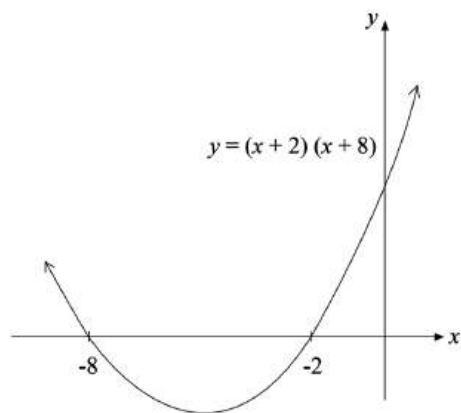


$$\begin{aligned}
 \text{a. } f(g(x)) &= \sqrt{(x+5)^2 - 9} \\
 &= \sqrt{x^2 + 10x + 16} \\
 &= \sqrt{(x+2)(x+8)}
 \end{aligned}$$

$$\therefore c = 2, d = 8 \text{ or } c = 8, d = 2$$

b. Find x such that:

$$(x+2)(x+8) \geq 0$$



♦♦♦ Mean mark 13%.

MARKER'S COMMENT: "Very poorly answered" with a common response of $-3 \leq x \leq 3$ that ignored the information from part (a).

$$(x+2)(x+8) \geq 0 \text{ when}$$

$$x \leq -8 \text{ or } x \geq -2$$

$$\therefore \text{Domain: } x \leq -8 \text{ and } x \geq -2$$